

Synthetic boundary layout sample

This is an artificial signal case, not a model result. The declared out-of-dose predictive criterion supports: log segment/weight norm, segment stable-rank fraction, segment effective-rank fraction, segment row-norm CV, segment top-1% row energy, cumulative/weight norm, cumulative stable-rank fraction, segment subspace overlap to AdamW, cumulative subspace overlap to AdamW. All nine features and all four folds are retained. Undefined folds are not pooled as available cases. This is prediction outside a learning-rate dose on the same task suite, not causal mediation, statistical significance or evidence of equivalence.

Geometry feature	RMSE reduction	Improved	Defined	Criterion	Residual ρ
log segment/weight norm	+0.0329404	4/4	4/4	supported	+1.000
segment stable-rank fraction	+0.0329404	4/4	4/4	supported	+1.000
segment effective-rank fraction	+0.0329404	4/4	4/4	supported	+1.000
segment row-norm CV	+0.0329404	4/4	4/4	supported	+1.000
segment top-1% row energy	+0.0329404	4/4	4/4	supported	+1.000
cumulative/weight norm	+0.0329404	4/4	4/4	supported	+1.000
cumulative stable-rank fraction	+0.0329404	4/4	4/4	supported	+1.000
segment subspace overlap to AdamW	+0.0329404	4/4	4/4	supported	+1.000
cumulative subspace overlap to AdamW	+0.0329404	4/4	4/4	supported	+1.000

Table 1: Every original weight-space predictor under four-fold leave-dose-index-out prediction. Decisions use exact MSE comparisons, not displayed RMSE. A missing correlation is undefined, not zero; baseline equivalence is distinct from an unidentified or numerically unresolved prediction.